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CIS4394 - Agentic AI

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Personal Health and Habit Coach

Problem Statement and Motivation

In today's world, many people struggle to build and maintain healthy habits. With the proliferation of algorithm driven apps like TikTok, Instagram, and YouTube, many people end up forgetting about their commitment to their new habits within a few days. Many basic habit trackers and reminder apps don't adjust to your feedback, or adjust when the user falls behind. There is a huge gap for something to help people stay on track, and an AI agent that helps keep track of your habits could help many people stay committed to their goals.

The target users for this agentic solution include students, working professionals, and others who want to improve their habits. These groups are usually people who are struggling with their schedules, and need a bit of coaching to reach their goals. The solution is not restricted to these groups, a solution like this could be useful for anyone who is looking to improve their lives and their habits.

Agentic AI is great for this problem due to the lack of personalization and immediate feedback that exists in solutions on the market right now. There is a lot going on when building a habit, including goal interpretation, plan/schedule creation, daily monitoring, pattern analysis, revisions of plan, and feedback. This type of complex workflow cannot easily be handled by a rule-based system, and a human cannot give as much focus or attention to detail. A human who specializes in helping you build habits would only be available to you during certain times of the day or week, and would likely have other clients. An agentic AI solution is able to handle complex workflows like this through multiple agents who handle different parts of the workflows and collaborate. An agentic AI solution will also be available at all times, while being much lower cost, and more customizable than a human coach would be.

Use Cases and Scenarios

In scenario one, a user wants to exercise three times per week and go to bed before midnight. The user would first tell the agent this goal. The Planner agent processes and interprets the user's goal. The Scheduler agent would create a realistic weekly habit plan that works around the user’s predefined schedule. The front-end then displays a schedule and daily habit checklist.

In scenario two, a user has failed to complete their habits for two weeks. The agent would recognize this, and address it by adapting for the user. A Tracker agent would have been collecting daily completion data, an Analyzer agent would identify patterns of missed days, and an Adjuster agent would modify the habits to adjust to the difficulty the user is having. Finally, a Motivator agent would generate supportive feedback for the user, and increase the frequency of reminders.

Initial System Design

This solution would include many agents, and they would all interact and share information.

- Planner- Process and plan out how to help the user build the desired habits.
- Scheduler-Would look at the user’s calendar and fit the new habits into open slots.
- Tracker- Keep track of if the user is completing tasks or not.
- Analyzer- Look at the Tracker agent’s data and look for patterns of incomplete habits.
- Adjuster- Adjust the habit frequency/difficulty based on the user’s ability to complete habits.

❖ Motivator - Send motivating messages to users and reminders to keep on track.

This agent would use multiple tools, APIs, and data sources. The agent would need access to a local database or file storage for habit logs, the user's calendar for scheduling, and access to send reminders through Gmail or push notifications. The front end will be built using Streamlit or a similar tool.

Feasibility and Risk

As is usual when working with AI, there are always risks. Hallucination is always a risk to watch out for, as the agent may generate an unrealistic habit plan. Users can also be a risk themselves, with disengagement by not logging habits or ignoring notifications. Another risk is trying to build too many habits at once.

Some constraints include access, privacy, and computation. If users are not willing to allow access to storage, their calendar, or reminders, then the agent will not be able to successfully help the user build habits. Privacy is also a possible constraint, but anonymizing data or storing it locally should avoid that. Computation may become a problem if the subagents take too much time/computational power to do their jobs.

Project Timeline

❖ Weeks 1-2: Proposal and System Design

❖ Weeks 3-4: Work on Front-End Implementation, due March 5th

❖ Week 5-6: Work on Multi-Agent Workflow Implementation, due March 12th

❖ Weeks 7-8: Evaluation and Testing

❖ Final agent due March 26th